

FPGA PCB project meeting notes

September 17, 2026 | BioScience Research Collaborative | 29 min 36 sec

Participants identified: Gerald, Howard, and Zitong. David and an FPGA firmware collaborator were discussed as interface collaborators.

Main outcome

Prioritize a compact FPGA board and a workable FPGA-to-KR260 interface for the eight-chip, approximately 4K-channel recording system. Howard and Zitong should work together to identify everything needed for power-up, configuration, control and data transfer before finalizing the hardware. Gerald identified the FPGA-to-KR260 connection as the main unresolved interface.

System and responsibilities

Recording chips → routing board → FPGA board → KR260 → PC

- **Chip and routing side.** The routing board brings chip signals to the FPGA board. Two detachable board-to-board connectors were discussed. Gerald will provide chip pin locations and reference designs; Howard and Zitong need to agree on the mating connections.
- **FPGA side.** Howard's main responsibility is the FPGA board. It receives and organizes chip data and forwards it over fewer signal lines. The two board designers should collaborate, especially on the less familiar FPGA support circuitry.
- **Downstream side.** KR260 was named as the downstream platform and called the "MCU" in the meeting. Coordinate its interface with David and the FPGA firmware collaborator. The intended control path comes from this downstream platform.

Agreed direction

- **Design tool.** Use KiCad for the PCB work. Check version compatibility when opening the shared files.
- **FPGA baseline.** The device discussed was Artix-7 XC7A35T, and Gerald agreed to use the referenced FPGA. Confirm the complete ordering code and package from that reference before finalizing the design.
- **Size and usability.** Minimize the board and support components, with a preference for a compact rather than elongated shape. Users should be able to power it on without manually reprogramming it each time.
- **Power and fabrication.** Place FPGA power circuitry on the FPGA board. Recording-chip power placement remains under review. JLC was the preferred fabrication/assembly option.

Technical discussion and open questions

- **Chip-to-FPGA pin count.** About 88 data signals and roughly 100–110 total signals including programming/control were discussed. These are estimates, not an approved connector pinout; the transcript contains inconsistent later numbers. Gerald's actual pin map must settle the count and direction of each signal.
- **FPGA-to-KR260 link.** USB-C was favored as a possible compact connector, with micro-HDMI as an alternative. Selection depends on the usable cable conductors and interface needs. No USB or HDMI signaling protocol was agreed. A downstream adapter board was also proposed.
- **Data aggregation.** Combining chip outputs into two to four data lines was a design question, not a demonstrated capability or finalized lane count. The interface also needs to accommodate control and programming. The team must determine what the hardware and firmware can support together.
- **Configuration and storage.** Local memory that loads the FPGA at startup was discussed as a way to avoid manual reprogramming after power-off. Memory selection, loading method, required pins and the downstream platform's role remain open. Configuration storage and recording-data buffering need separate requirements.
- **Power and board layout.** Required FPGA supplies and support components need a datasheet/reference-design review. Placement of recording-chip LDOs may change to reduce size. The dimensions quoted for the existing reference were approximately 23×23 for the FPGA and 60×50 for the board; these are not new-board targets. Final dimensions and layer count were not selected.
- **Future functions.** High-pass and low-pass filtering were mentioned as possible future requirements from Chong. They were not committed as immediate deliverables.

Questions to resolve in the next review

1. What complete FPGA part number and package will the design use, and what is the minimum circuitry needed to power, start, configure and operate it?
2. What is the exact chip-to-FPGA signal list, including data, timing, control and programming signals and their directions?
3. What data rate must the FPGA-to-KR260 link carry, and how many data and control lines does the chosen implementation require?
4. Which connector, cable and signaling scheme can both the FPGA firmware and KR260 support?
5. How will initial programming, automatic startup and later configuration updates work for the end user?
6. Where will recording-chip power regulation sit, and what dimensions and PCB layer count are practical?

Action items

Prepare the FPGA requirements and interface questions for the next discussion. Apart from that next-review expectation, individual completion dates were not set.

Research the minimum FPGA support design

Owner: Howard with Zitong

Bring a component/function list covering power, startup, configuration, programming, input/output and data transfer, with unresolved choices clearly marked.

Provide authoritative chip information

Owner: Gerald

Supply chip pin locations. Chip-board and older routing/LDO-board examples were shared during the meeting; the older routing example needs adaptation for eight chips.

Review references and align board interfaces

Owner: Howard and Zitong

Open the shared designs in a compatible KiCad version, reconcile the pin list, and agree on connector placement and mating connections.

Develop the FPGA-to-KR260 interface

Owner: Howard and Zitong with the FPGA firmware collaborator and David

Review available pins, data aggregation, control/programming needs and downstream support together. Gerald can join the discussion.

Finalize recurring-meeting logistics

Owner: Meeting organizer to confirm

Confirm the invitation and room for Wednesday at 4:00 PM. Clarify whether a Zoom link is needed.

Next meeting and other logistics

The group agreed on a recurring Wednesday meeting at 4:00 PM, after considering earlier times. Gerald expected to attend in person. The room, invitation owner and any Zoom arrangements still need confirmation. The next regular Wednesday is September 23, 2026, subject to the calendar invitation.

A possible ninth-floor desk was discussed, but availability and assignment were not confirmed.

Recording reference

BioScience Research Collaborative.m4a — recorded September 17, 2026 at approximately 1:19 PM Central; duration 29 minutes 36 seconds. The FPGA firmware collaborator's name needs confirmation.